

Justin Zhang

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TECHNICAL SKILLS

Skills: PID Control, CV, SLA/FDM 3D Printing, Machining/Manufacturing, DMM, Oscilloscope, DFM/DFMA, GD&T, PCB Testing

Languages: C++, C, Git, MatLab, Simulink, Python, RobotC, VHDL

CAD/CAM Softwares: SolidWorks, SW PDM, Upchain Cloud, SW Simulation, Ansys Discovery, AutoCAD, Fusion360, PrusaSlicer

EDA/Hardware Programming Softwares: KiCAD, Altium Designer, VESC, Siemens SIMATIC STEP 7

EXPERIENCE

Automation Designer - Mechanical (Co-op)

Sept. 2025 – Dec. 2025

ATS Corporation

Cambridge, ON

- Updated cell-level assemblies and station assemblies to reflect updates made on automation machines
- Created cell-level layout drawings for automation machines for use in internal design reviews and safety reviews
- Collaborated with machine builders to create solutions for problems discovered during the assembly phase
- Coordinated with vision and assembly teams to create guarding for vision station cameras and rotary table indexer
- Implemented various sensors and actuators such as pneumatic cylinders/slides, thru-beam sensors, and lighting for vision systems

Mechanical Engineering Co-op

Jan. 2025 – Apr. 2025

Electrium Mobility

Waterloo, ON

- Led mechanical sub-team and managed logistics for the electric longtail conversion kit project
- Assisted longtail conversion kit electrical subteam in wiring for the battery pack and kit
- Assisted in the design and fabrication of parts for the terms projects including go kart, custom RC, and onewheel
- Created mechanical workshops to teach members the basics of CAD design

Engineering Co-op (Mechatronics)

Sep. 2023 – Dec. 2023

RAB Design Lighting Inc.

Toronto, ON

- Redesigned housing using DFMA techniques and specified new LED Drivers and module to reduce cost by 30%
- Conducted failure analysis and identified root cause for failure on RMA lighting fixtures
- Completed quality control inspection of engineering and production samples
- Created dimension drawings for spec sheets and for machine shop production
- Generated installation instructions for end customer and contractors to assist in fixture installation

QA Engineering Lab Assistant Co-op Student

Apr. 2024 – Aug. 2024

Amphenol Corp Canada

Markham, ON

- Conducted various physical (IP, IK, Durability) and electrical tests (Low Level Contact Resistance, Dielectric Withstanding Voltage/Insulation Resistance) on electrical connectors to EIA, ASTM, ASME, and IEEE standards
- Operated and programmed specialized testing machines, used to conduct various environmental tests
- Updated 3D models and drawings to accommodate for changes in new injection molds
- Compiled test results into a QA Test Report (QTR) to be used as a certification document for the end product

PROJECTS

Autonomous Search and Rescue Robot | *PD Control, OpenCV, DFM/DFMA*

Sept. 2023 – Dec. 2023

- Co-designed and co-built robot which uses CV for line detection and PD control for drive control/path following
- Designed fully 3D-printed modular chassis for ease of assembly, compactness, and quick swapping of boards
- Programmed in Python using OpenCV library and state machine program structure

Injection Molded Steering Wheel Grips | *SolidWorks, Mold Design, 3D Printing/Print Finishing*

Mar. 2025

- Designed reusable 2-part injection mold for overmolded steering wheel grips in SolidWorks
- Used mold with Shore 70A urethane resin to create durable rubber grips for a go kart steering wheel

More info is available on justinz.ca if you would like to learn more about these projects and others!

DESIGN TEAMS

Electrium Mobility | *Team Lead - Mechanical*

April 2025 – Sept. 2025

- Leading student design team and advising on mechanical design for all aspects of the team

Electrium Mobility | *Project Lead - Electric Longtail Conversion Kit*

Jan. 2025 – Apr. 2025

- Co-led team of 13 members to create a conversion kit to convert a standard bike into an electric cargo bike
- Frame designed in SolidWorks with DFMA concepts and project budget in mind
- Completed finite element analysis (FEA) in SolidWorks Simulation to achieve a Safety Factor of 2